

Viva Questions and Answers (ASAP Brainwave Classification)

A) Basic Project Understanding

1. What is the main objective of this project?

- To classify emotional state from EEG signals using DREAMER dataset.
- We implemented both:
 - 4-class quadrant model ('Happy', 'Stressed', 'Depressed', 'Calm')
 - Binary models ('valence' low/high, 'arousal' low/high)

2. Why did you choose EEG-based emotion recognition?

- EEG captures brain activity directly, which can be useful for affective computing.
- It is a strong ML use-case combining signal processing + deep learning.

3. Which dataset did you use and why?

- DREAMER dataset ('DREAMER.mat').
- It is a known benchmark in EEG emotion recognition research.

4. What are valence and arousal?

- Valence: pleasantness (negative to positive).
- Arousal: activation level (calm to excited).

5. How do you map valence-arousal to 4 classes?

- High Valence + High Arousal -> Happy
- Low Valence + High Arousal -> Stressed
- Low Valence + Low Arousal -> Depressed
- High Valence + Low Arousal -> Calm

B) Data and Preprocessing

6. What is the EEG input shape used in your model?

- Raw window shape is '(14 channels, 256 samples)'.

7. Did you do manual preprocessing?

- No separate manual step is needed for user.
- Pipeline does preprocessing internally during training/inference.

8. Which preprocessing is used for 4-class path?

- Band Power Spectral Density extraction for theta/alpha/beta/gamma.
- Baseline removal.
- Final feature shape '(14, 4)'.

9. Why baseline correction is important?

- It reduces subject/session bias and highlights stimulus-driven changes.

10. How do you avoid data leakage?

- Scaler is fit only on training split per fold, then applied to validation data.

C) Model Architecture

11. Which models are implemented?

- 'EEGBiLSTMClassifier' for 4-class quadrant prediction.

- 'EEGNet' for binary valence/arousal prediction.

12. Why BiLSTM for quadrant model?

- It can model sequential dependencies in channel-wise feature sequences.

13. Why EEGNet for binary tasks?

- EEGNet is compact and designed for EEG signals, efficient for low-data settings.

14. What is confidence score in output?

- Probability of predicted class from softmax top value.

15. How is trial-level prediction done?

- By aggregating multiple window predictions:

- majority vote (with deterministic tie-break)

- mean probability strategy

D) Training Strategy

16. What split strategies are supported?

- 'cross_trial', 'cross_subject', 'loso', 'group_kfold_subject'.

17. Which split is most reliable for realistic performance?

- Subject-independent splits ('group_kfold_subject' or 'loso') are more realistic.

18. Why not trust single-fold accuracy?

- Single fold may be lucky/unlucky; multi-fold mean+std is more reliable.

19. Which metrics are reported?

- Accuracy

- Balanced Accuracy

- Macro F1

- (with mean/std across folds)

20. Why Balanced Accuracy and Macro F1 are important?

- EEG classes can be imbalanced; these metrics are fairer than plain accuracy.

21. What is early stopping and why used?

- Stops training when validation score stops improving.

- Prevents overfitting and saves compute time.

22. What optimization methods are used?

- AdamW optimizer

- ReduceLROnPlateau scheduler

- Optional focal loss, class weights, weighted sampling

23. What augmentations are used?

- Gaussian noise, channel dropout, and time masking (configurable).

24. How is best checkpoint selected?

- By composite score:

- $0.7 * \text{balanced_accuracy} + 0.3 * \text{macro_f1}$

E) Engineering and Deployment

25. How can a normal user run the project easily?

- Through 'run.py' commands:
- 'python run.py train'
- 'python run.py streamlit'
- 'python run.py api'
- 'python run.py evaluate'

26. What does Streamlit dashboard provide?

- Interactive model selection, payload testing, confidence/probability display.

27. What API endpoints are available?

- '/health'
- '/predict/features'
- '/predict/window'
- '/predict/trial'
- '/preprocess/demo'

28. How did you make project delivery-ready?

- Clean folder structure, removed cache/artifact clutter, added documentation.

29. What files are most important for reviewer?

- 'run.py', 'streamlit_app.py', 'training/train_eegnet_binary.py', 'inference/api.py', docs.

30. How do you ensure reproducibility?

- Fixed random seed + controlled split protocol + saved reports/checkpoints.

F) Performance and Limitations

31. Is this production-ready clinical system?

- No. It is a research/demo project for academic use.

32. Why accuracy may be moderate in EEG emotion tasks?

- High inter-subject variability, noisy signals, limited dataset size.

33. Is 75% guaranteed?

- No. It depends on split protocol and training conditions.

34. What is a realistic way to present results?

- Report protocol used, fold-wise mean+std, and limitations honestly.

35. What are current limitations?

- Moderate performance, subject variability, no real EEG device live streaming integration.

G) Improvement Questions (Future Work)

36. What improvements can be done next?

- Better hyperparameter search
- Subject adaptation/domain generalization
- Advanced time-frequency features
- Ensemble or multitask model

- Better artifact/noise handling

37. Can this support live prediction from EEG headset?

- Yes, with additional device SDK integration and real-time preprocessing pipeline.

38. Why camera-based live test is not equivalent here?

- This model expects EEG signals, not facial/video input.

39. How can confidence calibration be improved?

- Temperature scaling or calibration on validation set.

40. How can project be extended for publication-level work?

- Compare with more baselines, strict LOSO protocol, statistical significance tests.

H) Short 1-Line Answers (Rapid Fire)

- Dataset: DREAMER
- Input channels: 14
- Window size: 256 samples
- Binary thresholds: valence/arousal > 3.0
- Main DL models: BiLSTM + EEGNet
- Best validation selection: weighted score of balanced accuracy + macro F1
- Dashboard: Streamlit
- API framework: FastAPI
- Purpose: Academic research/demo

I) Demo Flow for Viva (Recommended)

1. Explain objective and dataset.
2. Show project structure quickly.
3. Run 'python run.py train' (or show existing report).
4. Show evaluation metrics JSON.
5. Run 'python run.py streamlit' and test sample payload.
6. Show 'python run.py api' and '/docs' endpoint.
7. End with limitations + future work.

Prepared for: College Viva / Project Defense

Project: ASAP Brainwave Classification